

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/284446202>

The Cultural Schema: Toward Conceptual Compatibility in Culture-Cognition Interaction Research

Chapter · February 2017

CITATION

1

READS

2,531

1 author:



Jacob Strandell

Swedish University of Agricultural Sciences

9 PUBLICATIONS 29 CITATIONS

SEE PROFILE

Some of the authors of this publication are also working on these related projects:



Culture-Cognition Interaction: Bridging Cognitive Science and Cultural Sociology [View project](#)



Governance, justice and resistance - on the way to a fossil-free welfare society [View project](#)

This is an updated online version of a dissertation chapter. Please cite the original dissertation using the following details:

Strandell, Jacob (2017) “The Cultural Schema: Towards Conceptual Compatibility in Culture-Cognition Interaction” in *Culture-Cognition Interaction: Bridging Cognitive Science and Cultural Sociology*. Doctoral dissertation, University of Copenhagen.

The Cultural Schema: Towards Conceptual Compatibility in Culture-Cognition Interaction Research

Dissertation chapter (University of Copenhagen)

Jacob Strandell

ORCID ID: 0000-0003-2984-9967

Abstract

A growing number of sociologists are now joining the interdisciplinary study of culture-cognition interaction, but broader engagement has been prevented by the vastly different vocabularies of cognitive and cultural theories, which creates an illusion of mutual incompatibility and irrelevance. The concept of ‘Cultural schemas’ can provide a conceptual bridge between these vocabularies, but the relationship between cultural and cognitive schemas requires further clarification to facilitate broader engagement and two-way compatibility. This paper reviews the schema concept and makes four interrelated arguments for conceptualizing cultural schemas and their relationship to cognitive schemas and other cultural conceptualizations. Cultural schemas should: 1) be conceptualized as a social analogue to cognitive schemas rather than as a subtype of mental patterns; 2) have the same conceptual role and structure as cognitive schemas in relation to other concepts, as an umbrella term subsuming other, more specific conceptualizations of culture; 3) be thought of as a supra-individual phenomenon with emergent properties, while still recognizing the neurocognitive microfoundations of culture; and 4) emphasize the asymmetric relationships between culture and cognition, and the disparate mental and social mechanisms involved as one shapes another. Finally, a multilevel analytical framework of this relationship is suggested based on established macro-micro models.

Keywords

Cognitive schemas, cultural schemas, cognitive sociology, internalization, culture-cognition interaction, reproduction

Acknowledgements

I want to thank Jakob Demant for supervision and encouragement, Stephen Vaisey for welcoming me to visit Duke University, and Naomi Quinn for her generosity and valuable feedback. I also want to thank everyone who has commented on past versions and drafts of the paper.

Introduction

Culture and cognition are two heavily intertwined concepts, describing phenomenon that are essential to just about any human behavior. This paper aims to provide a conceptual bridge between the theoretical frameworks of culture and cognition researchers by addressing an ongoing debate between cognitive sociologists. Sociologists have always made assumptions about the cognitive processes that transform culture into human action (Cerulo 2010; Danna 2014; DiMaggio 1997), any theory involving some form of *internalization* or *socialization* necessarily makes more or less explicit assumptions about learning, memory, and decision-making (Lizardo 2015). However, sociologists have despite fact that cultural processes rely on cognitive micromechanisms largely kept their distance from sciences studying the mind, thereby failing to integrate their insights and engaging in broader interdisciplinary conversations.

An increasing number of cultural sociologists are now urging their colleagues to engage with cognitive science, but sociology is arguably very late to the culture-cognition interaction party (Cerulo 2014b; Danna 2014; DiMaggio 2002; Lizardo 2014). At the turn of the 21st century, sociology was the only social science remaining outside of this immensely influential development of contemporary thought (Cerulo 2014a; Pitts-Taylor 2014). As psychologists, linguists, economists, philosophers, and anthropologists have joined in fruitful collaborations with cognitive scientists, sociologists have been remarkably impervious to cognitive science, “clinging to the sidelines, honoring—and in some cases strengthening—rigid intellectual boundaries” (Cerulo 2010:116). A central obstacle for sociologists to engage with cognitive science seems to be miscommunication and a mutual illiteracy of dissociated vocabularies, even in overlapping substantive fields of research (Abbe, Rentsch, and Mot 2009; Danna 2014; DiMaggio 1997; Pitts-Taylor 2014).

Due to the academic compartmentalization of culture and cognition research, many cultural sociologists still operate with an outdated notion of cognition (Lizardo 2014). This disconnect of frameworks and concepts is evident in how many sociologists use the term cognition in an everyday sense as synonymous to *thinking*—a misconception that gives the impression that cognition has nothing to do with emotions, behavior, attitudes, or intuition. Yet cognitive scientists view these as seamlessly intertwined with and dependent on cognitive processes. It is

only by reducing cognition to thinking that one can make sense of the notion of *non-cognitive skills* in educational sociology (c.f. Gutman and Schoon 2013), or the idea that an *affective turn* would be a move away from cognition (Leys 2011).

This mutual illiteracy leads cultural sociologists and cognitive scientists to misperceive each other's work as incommensurable and irrelevant (Danna 2014; Lizardo 2014), yet contemporary cognitive science confirms and supports many insights of cultural sociology (DiMaggio 2002; Lizardo and Strand 2010; Vaisey 2009). This leads sociologists to miss out on the great advances that are now possible in our understanding of culture-cognition interaction. The absence of sociology is evident as emerging new subdisciplines, such as cultural psychology and cultural neuroscience, recognize cultural anthropology as an essential influence while omitting cultural sociology (Chiao et al. 2010; Valsiner 2014). For sociology to remain relevant, much less strengthen its position, in 21st century social science, its conceptual frameworks must be aligned with overlapping disciplines (Lizardo 2014; Pitts-Taylor 2014). There is much to be done to span the conceptual chasm between culture and cognition researchers (Danna 2014), but some such translation work has already begun (c.f. Cerulo 2010; DiMaggio 2002; Lizardo and Strand 2010; Vaisey 2009).

Cultural schemas have been suggested as a suitable base concept for bridging culture and cognition frameworks (DiMaggio 1997), and have seen increasing use in empirical research emphasizing culture-cognition interaction (c.f. Abbe et al. 2009; Baumann and Ho 2014; Baumann and de Laat 2012; Blair-Loy 2001; Brubaker, Loveman, and Stamatov 2004; Cerulo and Ruane 2014; Frye 2013; Fallin Hunzaker 2014; Vaisey 2009). This concept is potentially powerful because of its close semantic and structural resemblance to cognitive schemas, a base concept of cognitive science. This similarity makes the concept accessible for two-way compatibility and enables ready access to insights from cognitive schema research to bolster our understanding of culture in action. However, sociological definitions of cultural schemas are scarce, contradictory, and often vague regarding the relationship between cultural and cognitive schemas.¹ Furthermore, the relationship between cultural schemas and other

¹ Other conceptualizations have, however, been developed by cross-cultural communication researchers (Nishida 2005; Sharifian 2003) and cognitive anthropologists (D'Andrade 1995; Quinn and Strauss 1997).

established sociological conceptualizations of culture is unclear, which may be an obstacle to interdisciplinary engagement.

This paper reviews the utility of the cognitive schema concept for sociological phenomenon and offers a conceptualization of cultural schemas with two-way compatibility intended to promote mutual accessibility, analytical utility and attractiveness to culture-cognition interaction researchers coming from both sides of the fence. This conceptualization should therefore be designed to facilitate the cumulative use of previous research from both the cognitive sciences and cultural sociology, as well as to minimize *psychophobia* (D’Andrade 1992), the fear of psychological reductionism. Therefore, the aim has been to avoid attributing primacy to either culture or cognition, or subsuming one under the other.

The Cognitive Schema

The cognitive schema is well-established as a basic unit of mental information processing in contemporary cognitive science, here defined as *a set of associated features representing a recurring pattern in the environment*. The reader should be aware that *features* and *representations* are used in the broadest possible way here. Representations should therefore not be understood as models, images or symbols, but as a cognitive correspondent to a phenomenon in the external environment. All imaginable ‘things’ in the world that can be described as a pattern of related features—including objects, people, events, abstractions, emotions, interactions, roles, and places—can be represented by a schema consisting of the features common to that entity (D’Andrade 1995; Mandler 1984; Quinn and Strauss 1997). A bird, for example, can be represented by the set of features that commonly co-occur whenever we experience, talk, or think of birds, such as them being animals and having wings, feathers, and a beak (see Figure 1). These schemas are in turn used to organize, process, and respond efficiently to similar experiences.

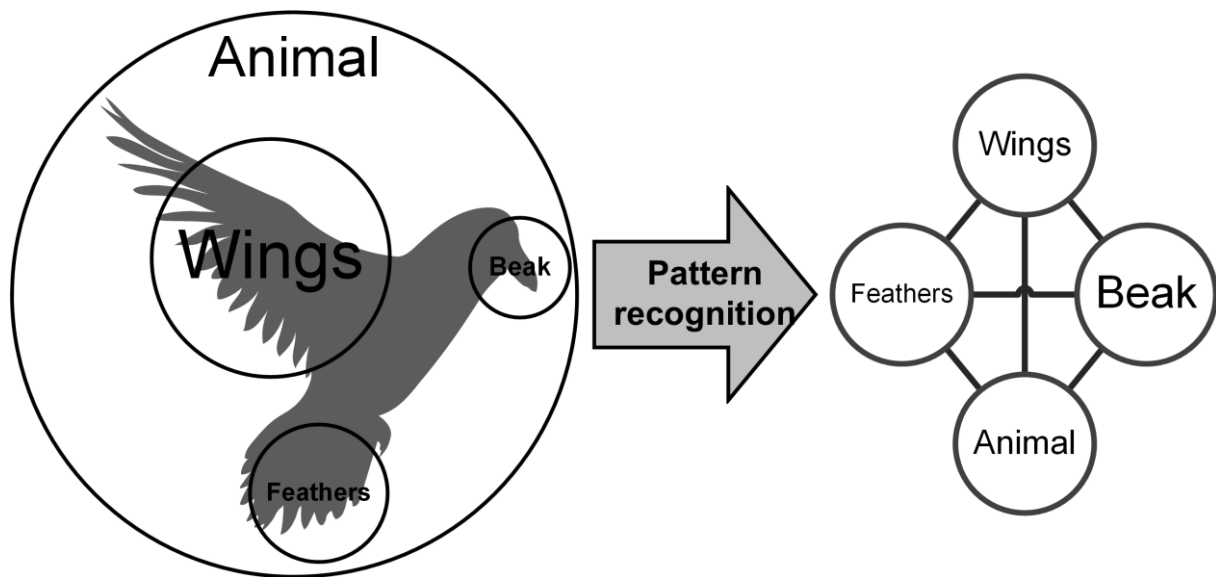


Figure 1: Pattern recognition using a bird schema abstracted from features common in most birds

The nodes in a schematic model, such as the model depicted in Figure 1, represent a distinct pattern of neural clusters connected by associations of varying strength (c.f. Conrey and Smith 2007). The strength of a connection increases with repeated use, increasing the likelihood that the schema will be used again. Consequently, application and learning of schemas occur simultaneously. Whenever a pattern of neural connections activates, it is reinforced and becomes more salient and accessible in the future (Mandler 1984). People, therefore, learn all kinds of things automatically, unintentionally, and often unconsciously, when exposed to recurring patterns (Bargh, Chen, and Burrows 1996; Bargh 1997). This process of *automatic internalization* can explain how and why people acquire such a broad set of cultural knowledge, including norms, values, and ideals that they do not actually endorse (e.g. ‘racism without racists,’ see Shepherd 2011). People extract abstract schematic patterns from salient co-occurrences and thus implicit attitudes are formed and maintained, which can explain some forms of structural discrimination, such as when people associate skin color or gender with positive or negative traits reproduced out of prejudice or structural conditions. The famous doll studies by Clark and Clark (1950) are excellent examples that show how small children have come to associate the skin color of otherwise identical dolls with positive or negative traits.

Even though it is constantly and automatically ongoing, schematic pattern recognition and learning is not a matter of passive internalization of rigid predetermined categories. Instead of simply classifying things as within or outside a category, objects are recognized by their degree of fit with previously learned schemas, and they are assigned a graded membership status, as belonging more or less to a category (c.f. Rosch 1973). This reflects the flexibility of real-life cognition. For example, a three-legged dog can instantly be identified as a dog because, despite missing a common feature of dogs (four legs), it stills fit better with a dog schema than any other schema. Variations of an abstract prototypical schema can occur by adding or removing features from the schema, creating more specific or concrete examples (see Figure 2).

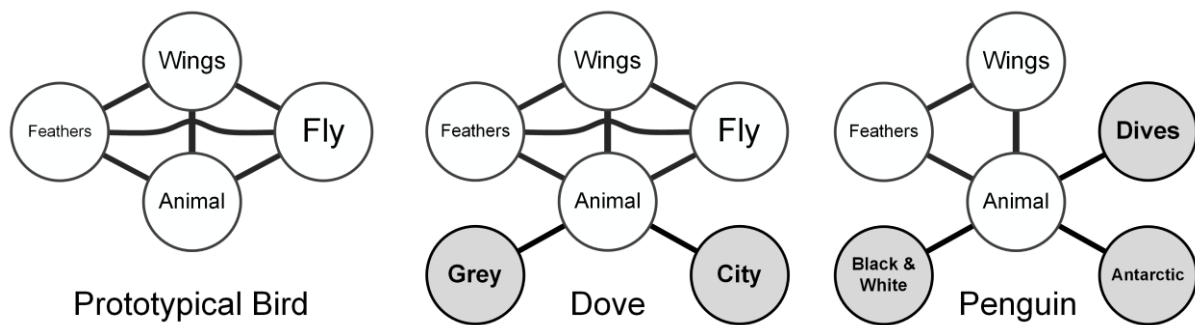


Figure 2: Variations of a bird schema, adding and removing features of the abstract and generic prototype. As the dove schema resembles a prototypical bird more closely than the penguin, it will also be more accessible when thinking of concrete examples of birds

This is likely another microfoundational mechanism for many socio-cultural phenomena, such as stereotypes and norms, as people abstract prototypical examples of social groups to construct generic schemas that are later applied to individuals based on salient features (Shepherd 2011). Prototypicality and graded membership also explain phenomena such as when people consider immigrants from countries similar to their own as ‘less immigrant’ than those from countries that differ further.

People attain abstract schemas by identifying features that are relatively stable across variations of a pattern in different instantiations and situations, and disaggregating these central features from idiosyncratic ones. This extraction of general features explains why studies show that “people actually harbor abstract cultural schemas about who ‘men’ and ‘women’ are, despite

the fact that no one has ever met someone who is a man or a woman without also occupying many other [intersecting] identities at the same time" (Ridgeway 2006:8). These abstract but culturally recurring schemas, such as roles or identities derived from recurring experiences, allow for efficient everyday navigation of the social world and interaction with people who share similar cultural experiences (Berger and Luckmann 1966; Quinn and Strauss 1997). In experiments where people retell stories, cultural transmission research shows how lost information is gradually replaced by stereotypical assumptions from the storyteller's own cultural experiences (Bartlett 1938; Fallin Hunzaker 2014). To some extent, this mechanism accounts for the cultural stereotypes and implicit presumptions that persist.

Abstracted generic schemas are modular and transposable,² and are used to fill in the blanks of missing information or anticipated actions and events. A restaurant-schema may, for example, provide generic expectations of the roles, behavioral scripts, organization, and expectations of services that most restaurants share (Schank and Abelson 1975). We usually do not need to learn anew that there will be food, menus, tables, waiters, cooks, customers, an economic exchange, and so forth. This cognitive ability provides a micromechanism for the sociological intuition that people attain cultural frames that enable or constrain future interpretations and actions (Bourdieu 1990; Goffman 1974; Laclau and Mouffe 1985).

While schemas account for some stability of cultural reproduction, abstract transposable schemas are also highly flexible and allow us to identify and understand entirely new experiences by piecing together hybrid schemas to match the experience as close as possible (Bourdieu 1977; Sewell 2005; Shepherd 2011). The process of recognizing patterns and learning from them is a process of dynamic construction, where people make use of their individual learning histories to creatively make sense of the world (Oyserman 2015; Ridgeway 2006). Because schematic interpretation is underdetermined and subject to more or less dynamic interpretations, schematic reproduction is not static, and the transposability of modular schemas is a mechanism of potential structural change (see Sewell 2005 for an extended argument).

²In cognitive science, 'generalizable' is a more common term but, as Sewell (1992) argues, generalizability implies further abstraction, while transposability simply refers to a move from one domain to another.

Cognitive schemas are modeled after neural networks to be commensurable with neuroscience and capable of the efficiency and flexibility of actual cognition (Smith 1996; Smith 1998). But while they model neural activity, schemas are not made up of physical neurons. Instead, a schema is a *pattern of activation* across configuration of neurons that fire together in a vast network of potential combinations (c.f. Conrey and Smith 2007). Therefore, schemas are not actually 'stored' as bounded units in the brain, but are instead enacted out of a near infinite number of potential activation patterns (Smith 1996, 1998). As people experience things, schemas activate and corresponding neural connections become reinforced, but what is 'stored' by this is the *tendency* towards reconstructing a *similar* schema, and not a distinct unit of knowledge (Mandler 1984).

Potential schematic patterns are superpositionally stored in neural networks and therefore overlap and blend into each other (Clark 1998; see Figure 3). Which pattern of activation is formed out of all potential combinations is determined by a domino-like effect of activation 'flowing' across the network through more reinforced paths, as well as by input points, such as sensory stimuli, other thoughts, or emotions (Collins and Loftus 1975; McClelland, Rumelhart, and McClelland 1986). Therefore, a schema is the outcome of a combination of i) associations differentially reinforced by experience; ii) input patterns from stimuli; and iii) intermediate schematic processing (e.g., a perception-schema activating an associated action-schema). All these components are subject to constant change, which means that each enactment of a schema will differ slightly from previous versions (Mandler 1984). As the specific outcome schema is not there prior to the activation process, and its shape is underdetermined by the neural network itself, schemas are always (re-)constructions.

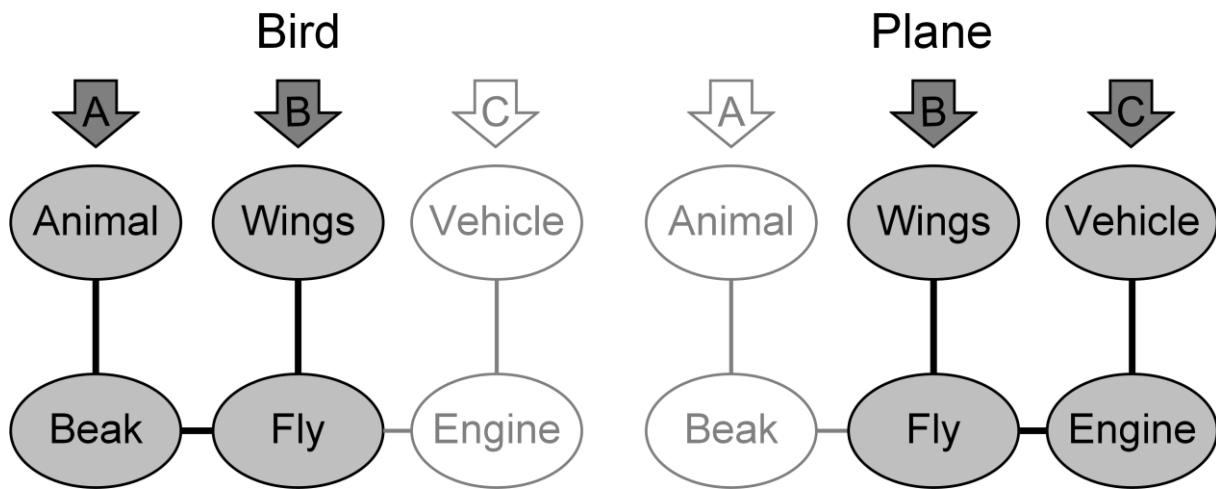


Figure 3: Activation flow from different input points creating different activation patterns (schemas) of overlapping nodes in the same physical network

Because schema activation depends on input cues, schemas are highly context dependent, and consequently so is learning (Bayer, Campbell, and Ling 2015; DiMaggio 1997; Shepherd 2011). Schemas are learned in association with the contexts where they are commonly experienced, including physical environments, social instructions, relations, or even abstract domains. Therefore, some schemas are more salient and more easily activated in certain contexts, and contexts—such as the physical environment, the presence of cultural symbols, or the sequential ordering of discourse—provide input influencing the schemas people use to perceive, understand, and react to a situation (Bargh 1997; Bayer et al. 2015; Cerulo and Ruane 2014; Shepherd 2011). For example, robbery may be more likely to come to mind than internet piracy when discussing ‘crime,’ but if the contextualizing topic is both ‘online’ and ‘crime,’ internet piracy might be more likely to come to mind (see Figure 4). Because of this context dependence, socio-cultural contexts can be said to operate as scaffolding devices for cognitive processes (Lizardo and Strand 2010).

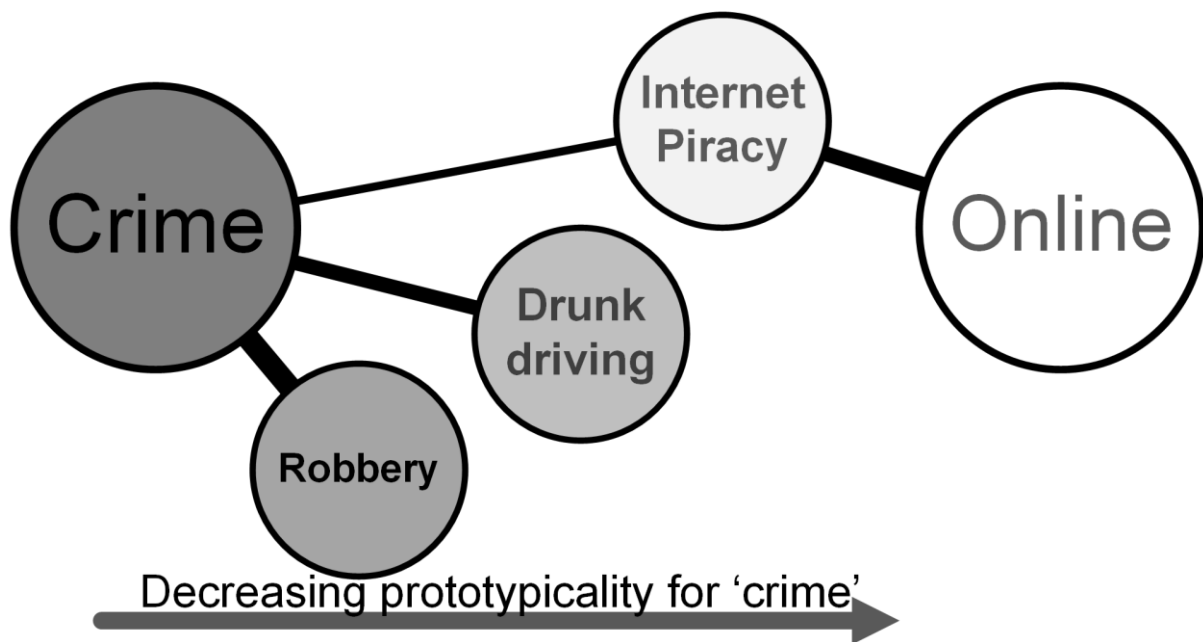


Figure 4: The influence of topic context on schema activation

Schematic processing is far from simple stimulus-response associations. Despite the fact that the microfoundation of cognitive schemas is neural networks, which alone are little more than on-off switches, the sheer size of such networks make schemas capable of processing complex cultural logics (c.f. Evans 2015; Shokouhi and Moghimi 2015). With an average of 86 billion neurons, the brain is a network of staggering complexity, which can represent complex abstract logic by utilizing multiple intermediate layers of interconnected nodes, as illustrated in Figure 5. Consequently, cognitive schemas can, despite their relatively simple mechanisms, constitute the microfoundation for abstract cultural logics, such as discourses, and not only direct associations. This means that schematic processing can explain how people automatically learn and apply cultural logics through repeated exposure.

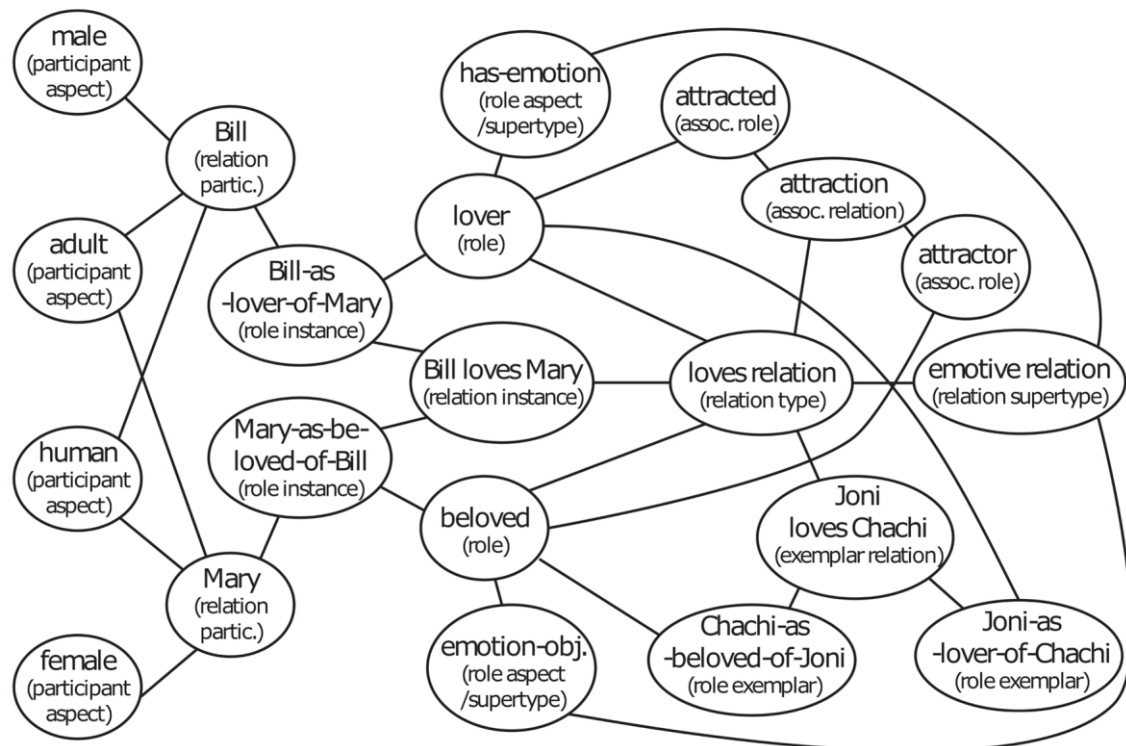


Figure 5: Example of a complex schema of a love relationship (reproduced with author’s permission from Evans 2015)

The cognitive schema is a powerful, well-developed, and well-researched concept that can provide unique explanatory value to cultural analysis. Schemas explain how people organize, process, respond to, and learn from experiences, including tacit, material, behavioral, or unconscious aspects. Schemas are both representational and processing mechanisms, and they are motivational as actions and emotions are learned in direct association with perceptions and interpretations (D’Andrade 1992; Quinn and Mathews 2016). An important implication of the fact that cognitive schemas are both representational and processing mechanisms, is that culture shapes not only the content of cognition but also its mechanism (Smith 1996). In addition to providing a highly detailed account of the micromechanisms of cultural phenomena such as norms, stereotypes, and frames, the cognitive schema concept is also potentially compatible with many established research traditions, including constructionist research and analyses of cultural logics.

‘Cultural’ Schemas?

For cultural schemas to serve as a conceptual bridge between the cognitive scientist’s vocabulary of neurons and networks and the cultural sociologist’s concepts of narratives, imaginaries, and repertoires, we need to have some idea of what we mean by culture and how culture differs from cognition. Culture is difficult to define—Kroeber and Kluckhohn (1952) once classified 162 different nuances in definitions of culture—but in this context it is sufficient to roughly distinguish how cultural sociologists use the term. Cultural sociologists have developed a fairly sophisticated idea of culture over the last few decades (c.f. Alexander 2003), which, I will argue, differs substantially both from many other social science disciplines and from the common, everyday use of the term.

In intercultural communications research, for example, Nishida (2005) conceptualized cultural schemas as *nationally shared cognitive schemas*, such as schemas of traditional American birthday celebrations. Thinking of cultural schemas as shared cognitive schemas is also common in cognitive anthropology (c.f. Quinn and Strauss 1997). This will be discussed in more detail further on, but for now we will focus on the idea that culture is national. This is similar to a common, everyday use of the term culture, as a set of *ethnic or national traditions*, including “the beliefs, customs, arts, etc., of a particular society, group, place, or time” (Merriam Webster 2015). This is a relatively narrow view of culture, because only certain types of practices or ideas are deemed cultural (e.g., traditional, moral, or artistic) and only certain clearly identifiable societal-level groups are said to have culture (e.g., nationalities, ethnicities). This narrow view is also a common presumption in other social science disciplines, such as the comparative psychological research contrasting Eastern and Western cultures (c.f. Miyamoto, Nisbett, and Masuda 2006; Oyserman, Kemmelmeier, and Coon 2002).

Viewing culture as national-ethnic traditions makes sense in intercultural communications research and comparative research, but it differs sharply from the broad view common in cultural sociology. In this view, culture is not constrained to certain types of behavior or groups, but instead is potentially applicable to all kinds of meanings or practices that are socially reproduced. While there is a good point in studying socially reproduced meanings and practices in car factories or medical diagnoses, this omnipresence of culture of course also makes it very difficult to specify. Goodenough (1957:167), one of the forerunners of cognitive anthropology,

famously defined culture as broadly as "*whatever it is one has to know or believe in order to operate in a manner acceptable to [a society's] members.*" Yet even this broad definition, which has seen much productive use in empirical research, does not cover the research object of cultural sociology. People sometimes behave in socially unacceptable or dysfunctional ways with cultural origins; the research object of cultural sociology is not limited to norms or beliefs, but also includes, for example, institutionalized practices; moreover, subsocietal groups arguably have different and sometimes conflicting cultures (e.g., political communities, organizations, lifestyles, and subcultures).

It seems, therefore, that a broad definition of culture is necessary, where culture potentially could encompass everything. Yet when we say that we study culture we nonetheless study some certain kind of phenomena with certain characteristics. Despite being an unfashionable citation, we can draw on Parsons (1951) to establish a few defining characteristics of what we call culture: that which is *socially transmitted, learned, and shared*. All three are necessary; while diseases are transmitted and shared, they are not learned, and learning on a deserted island is not socially transmitted or shared, but learning to be a productive member of a car factory involves the social transmission of shared skills. While these three characteristics define culture more specifically, culture essentially refers to the social reproduction of patterns in meaning or practices. This definition is sufficiently broad, yet it specifies culture as that unique human ability to distribute cognition, for example, in the scientific community, and to convey supra-individual knowledge from one generation to another.

There is a larger point to distinguishing a broad use of culture from a narrow use than simply developing a concept of cultural schemas that is suitable for culture-cognition interaction researchers. As with diverse uses of 'cognition,' different ways of talking about 'culture' can produce misunderstandings and, in a scientific context, an illusion of irrelevance. Much like cognitive science may appear irrelevant to sociologists who equate cognition with thinking, cultural sociology may appear irrelevant to cognitive scientists if they have a narrower everyday idea of culture. Arts and ethnic traditions may be of little importance to cognitive scientists studying decision making or interpretation processes, yet ideas, knowledge, and behavioral scripts of a social origin are highly relevant. Therefore, establishing a conceptualization of cultural schemas referring to culture in the broad sense is important, because it emphasizes the

social origin of many more cognitive schemas than just those representing American birthday parties. This highlights the utility and importance for cognitive scientists to recognize and engage with sociological research for understanding most everyday cognitive processes.

A Cultural Analogue: Having and Eating the Cake

Researchers generally agree that culture has a profound influence on individual cognition. Culturally shaped cognitive schemas provide ready cognitive solutions to recurring practical and social problems, provide people with frameworks for attributing causation to human actions, and knowledge about how the social world works. These shared, transposable, abstract, and prototypical pieces of information, consciously or unconsciously fill in the blanks in our perceptions and interpretations of the world (c.f. Fallin Hunzaker 2014; Martin and Desmond 2010; Quinn and Strauss 1997; Shepherd 2011).

Despite an agreement on the importance of culture, however, cognitive sociologists disagree on the relationship between culture and cognition (see DiMaggio 2002; Lizardo 2007, 2014, 2015; Sewell 2005; Strydom 2007; Turner 2007b). Some researchers argue that we should maintain the traditional ‘culturalist’ top-down perspective where culture is treated as a large-scale supra-individual phenomenon external to individual minds, while others argue that we should take a reductionist, ‘embodied’ or ‘individualist,’ view where culture is effectively reduced to interacting individuals. These different views on culture-cognition interaction needs to be resolved if the goal is to develop a conceptual framework integrating culture and cognition research.

The two views on culture-cognition interaction are also tied to two different ways of using the notion of cultural schemas. As illustrated in Figure 6, one conceptualization treats cultural schemas as a certain type of cognitive schemas, and thus as a micro-level phenomenon within individual cognition (Model A). The alternative to this (Model B) is to use cultural schemas to refer to a macro-level phenomenon, that is to socially recurring patterns, rather than to mental representations of those patterns (which in model B would be referred to as *cognitive* schemas).

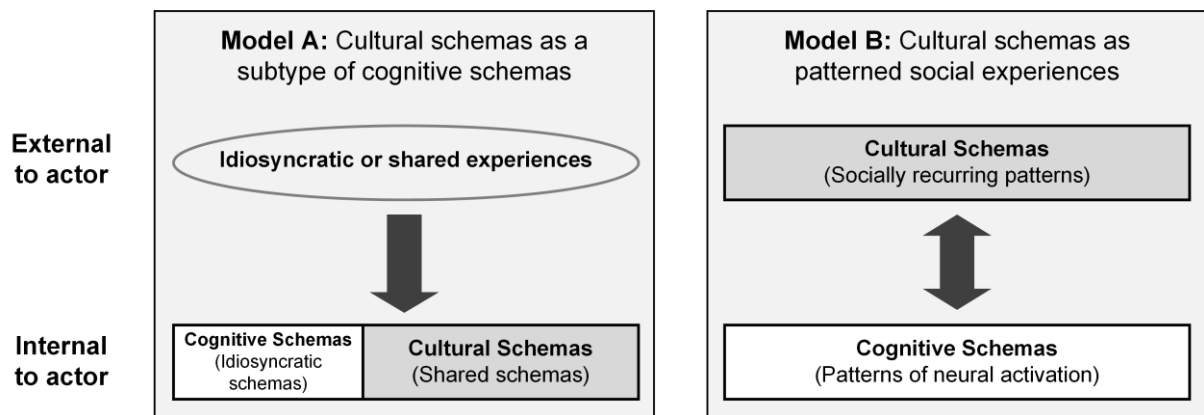


Figure 6: Cultural schemas can be conceptualized as a subtype of cognitive schemas, or as a supra-individual (social, external, and emergent) concept analogous to cognitive schemas

For the purpose of facilitating interdisciplinary exchange, this paper argues for the value of Model B and suggests that we should use cultural schemas as a macro-level analogue to cognitive schemas, rather than as a subtype of cognitive schemas. There are three main arguments for this, all tied to the goal of accessibility and utility as a bridging concept. A supra-individual conceptualization of cultural schemas: i) is better suited to the sociological research object: culture in the broad view, ii) has better compatibility with most sociological theories, and iii) is not incompatible with a strong emphasis on individual cognition as the microfoundation of culture if treated as an analytical concept.

A key problem issue with conceptualizing cultural schemas as a subtype of cognitive schemas (Model A in Figure 6) is that it makes little sense for cultural sociologists to make a distinction between personal/idiosyncratic schemas and cultural/shared schemas (such as in Nishida 2005 and Quinn and Strauss 1997). With a broad view of culture, very little is idiosyncratic or personal; just about all concepts, perceptions, thoughts, and practices are at least *influenced* by culture, with few exceptions. That is, our ideas, desires, and behaviors are largely socially learned, transmitted, and shared—including things such as highly personal memories, which are both originally experienced and later reconstructed using schemas heavily shaped by culture. Therefore, if cultural schemas are used as a type of cognitive schema all cultural schemas would per definition be cognitive schemas, and just about all cognitive schemas would necessarily be cultural schemas. Such a near-complete conceptual overlap makes the analytical

and theoretical utility of making a distinction between the two questionable in the first place. Even if we are talking about different degrees of cultural influence, both terms would refer to 'more or less culturally shaped mental patterns'. It seems to follow that treating cultural schemas as supra-individual provides more analytical value, because it describes something different which, I will argue, we need to have a concept for anyway, while remaining highly compatible with the cognitive schema.

There are, however, strong objections to conceptualizing the cultural schema as a supra-individual macro-level concept. Many culture researchers inspired by the cognitive sciences have recently argued against 'traditional' supra-individual views of culture, in favor of an 'individualist' view where culture refers to more or less shared cognitive schemas built out of interaction and shared experiences (c.f. Lizardo 2007, 2015; Nishida 2005; Quinn and Strauss 1997; Turner 2007b). The main objection of these researchers appears to be an aversion to what Lizardo has called "ontologically spurious anti-cognitive pseudo-objects" (2014:988), or what Turner (2007b) refers to as mysterious 'collective objects.' Looking closer at their arguments, it becomes clear that what Lizardo and Turner are objecting to are primarily the reification of macro-level analytical concepts, that is the treatment of things such as cultural schemas as 'ontologized units' existing only in 'the sociological ether.' This is a valid and serious concern. If cultural schemas are considered to be supra-individual objects, existing out there independently of cognitive processes we risk obfuscating the cognitive micromechanisms of culture in a way which only maintains the intellectual distance between cultural sociology and cognitive science.

However, there are two unwarranted assumptions in the critique against supra-individual culture concepts, which are that i) analytical concepts must have an independent ontological substance to be meaningful, and that ii) working with higher-order abstractions necessarily obscures their microfoundation. I will deal with both of these assumptions, and argue that the larger problem with compatibility with cognitive science is not the sociological use of abstractions, but the use of abstractions with disciplinary specific jargon.

It should be made clear here that a supra-individual conceptualization of cultural schemas is intended to be an *analytical concept* and not an *ontological object*, in the sense of Turner's (2007b) 'collective objects' and 'ontologized units.' As an analytical concept, cultural schemas

are far from mysterious—they are what they claim to be and nothing else: a way to refer to recurring social *patterns* observed by researchers or lay people. In other words, the notion of ‘cultural schemas’ is a shorthand way of talking about phenomenon emerging from complex processes, as sociologists have always done, but in a way that is much more compatible with the language and insights of cognitive science.

Is there a problem inherent in using abstractions to refer to constellations lower-order objects? No. All sciences routinely rely on abstractions to refer to emergent phenomena without for that reason disregarding their ontological microfoundations. Doing so is both an analytical shortcut and a way to highlight macro-level properties emerging from interacting micro-level phenomena (Kaidesoja 2013). It is, for example, common in the biological sciences to refer to complex entities such as ‘organs’ or the ‘nervous system,’ and to analytically treat these as emergent objects, without necessarily taking the detour of detailing the cells and molecules they consist of. This is analytically efficient since being able to talk about how one macro phenomenon influences another, such as how the nervous system interacts with certain organs, saves biologists the need to explain how ion channels open or neurotransmitters are released in every single neuron along the way. Yet, despite referring to emergent entities because they are analytically efficient, the biological sciences do not ignore micromechanisms and still rely largely on mechanistic explanations. Why could this not be the case in the social sciences?

We have a similar situation when studying culture-cognition interaction. Yes, it is true that cognition is the microfoundation of culture, and that there are no actual ‘collective objects’ out there external to the minds of individual people, but that is beside the point of an analytical macro-level concept. Turner (2007b), for example, refers to postmodernism as one of the reasons why sociology lagged behind other social sciences in engaging with cognitive science. This may be true, but postmodernism is an analytical construct and, in the conceptualization advocated here, a cultural schema (or a collection of such). Cultural ‘individualists’ must agree that it is convenient to be able to talk about postmodernism despite the fact that postmodernism lacks ontological substance and is simply an emergent phenomenon reconstructed post hoc by social theorists. We could, of course, theoretically trace everything that postmodernism is through millions of individual minds in order to avoid using a supra-individual construct, but that would be incredibly inefficient.

A reviewer of an earlier draft of this paper asked if an 'individualist' shared-cultural-schemas model of culture is not simpler than one that involves on higher-order abstractions. The reviewer went on to ask: is there "a cultural schema of an apple pie" or just "what I think of and what probably most people think of when they think of apple pie"? There is a misunderstanding here, as the reviewer jumps from cultural schemas as higher-order abstractions, to talking about them as ontological objects. Since I argue for the cultural schema as a higher-order abstraction with analytical efficiency, I would say that "a cultural schema of an apple pie" and "what I think of and what probably most people think of when they think of apple pie" are the same thing. As an analytical concept, the cultural schema simply refers to the fact that people in a certain culture reproduce consistent apple pies, and that their shared experiences of all these apple pies have certain causal effects. This is of course handled through the establishment of *similar* cognitive schemas (although not fully *shared*), but the presence of similar cognitive schemas among a group of people does not neglect the utility of a higher-order abstraction to describe the cultural phenomenon that all these people experience. To answer the reviewers first question, I would say that it is much simpler and more analytically efficient to talk about "a cultural schema of an apple pie" than "what I think of and what probably most people think of when they think of apple pie."

One might object to the earlier comparison with the biological sciences by saying that hearts and livers nonetheless have ontological substance, whereas cultural schemas do not. However, even advocates of an 'individualist' view of culture-cognition interaction rely on some form of abstractions that are not actually ontological objects. They often seem to forget that 'practices,' or even 'cognitive schemas,' are no more ontological solid than, say, 'discourses.' Yet such analytical concepts are treated as unproblematic objects. As previously discussed, in the cognitive sciences the cognitive schema is used to refer to a *pattern* emergent from firing neurons and *not to the physical neurons themselves* or a set of connections in a neurological network (Mandler 1984; Smith 1998). Cognitive schemas are consequently used much like a discourse analyst uses discourse: to refer to an analytical unit of observed tendencies, rather than to something that exists in an independent or solid form out there. Just as cognitive schemas emerge from interacting neurons, cultural schemas emerge from socially coordinated actions of individuals with similar cognitive schemas (c.f. Sharifian 2003).

We Find Many Names for Those we Love

A plethora of concepts has been developed to label cultural units, or different types of socially recurring patterns, resulting in different theoretical vocabularies that unfortunately have contributed to the internal fragmentation of sociology (c.f. Brante 2001; Lizardo 2014). Researchers may talk about discourses (Foucault 1972; Laclau and Mouffe 1985), practices (Bourdieu 1990), toolkits (Swidler 1986), scripts (c.f. Cerulo and Ruane 2014), value systems (Parsons 1951), models (D’Andrade 1992; Quinn and Strauss 1997), frames (Goffman 1974), or cultural conceptualizations and categories (Sharifian 2003), to name just a few. These are of course not the same, but despite emphasizing different aspects of culture, they share a common denominator in that they all refer to socially recurring patterns. What cultural sociology needs from a concept such as cultural schemas is, therefore, not yet another cultural unit, but rather an overarching abstract concept, relating these specific conceptualizations to each other as well as to cognitive schemas.

A key strength of the cognitive schema is that it is an umbrella concept that encompasses many different concepts for different types of cognitive patterns and integrates them into a commensurable framework. This capacity is something that the cultural schema should seek to emulate, both for the potential to better integrate cultural conceptualizations and because it would mirror the conceptual role of cognitive schemas, making them commensurable. In the cognitive sciences, the schema concept does not belong to any single discipline, theory, or field, but instead is a part of a shared conceptual vocabulary, integrating research from a diverse range of disciplines (Mandler 1984). As a generic concept, it subsumes more specific cognitive representations—such as image schemas, role schemas, evaluative scripts, or the self-concept—as variations of schemas, subject to the same basic structure, processes, and mechanisms. The cultural schema could fulfill a similar, corresponding, role for cultural theory.

As a conceptual analogue to cognitive schemas, the cultural schema should also be used as a high-level abstraction and an umbrella concept, a shared base concept referring simply to recurring cultural patterns of one type or another (c.f. Johnson-Hanks et al. 2011). Imitating the conceptual role of cognitive schemas would make cultural and cognitive schemas more

commensurable with each other (see Figure 7). It would also provide cultural sociology with a less esoteric concept for communicating research outside of the discipline. In addition, the umbrella model could also lessen resistance against a ‘cognitive turn’ from cultural sociologists invested in one of the established traditions, since the cultural schema concept is used as a conceptual adapter rather than a replacement for established concepts.

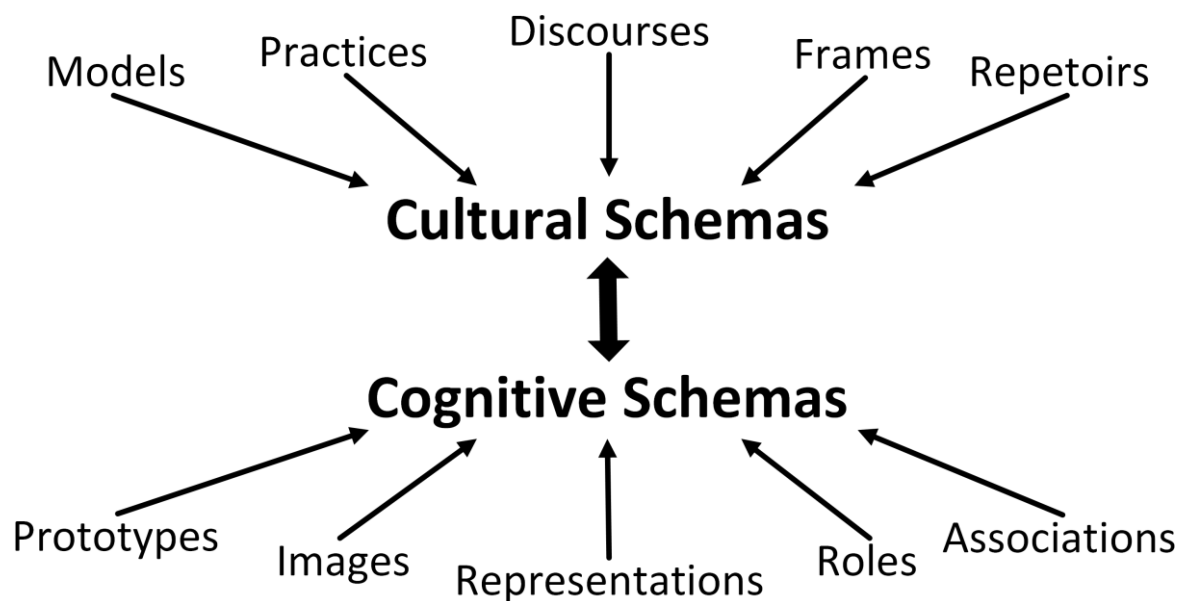


Figure 7: Conceptual map of the relationship between cultural and cognitive schemas used as umbrella concepts for other conceptualizations of culture and cognition

In contrast to the umbrella model, cultural schemas have sometimes been treated as yet another concept as other cultural conceptualizations, such as, for example cultural models, scripts, and categories (Cerulo and Ruane 2014; D’Andrade 1992; Quinn and Holland 1987; Sharifian 2003). However, cultural schemas are arguably better thought of as umbrella concepts on a higher level of abstraction, in part to imitate the role of cognitive schemas in the cognitive sciences, but also because schemas are more abstract and generic than models, scripts, or categories, which are all schematic. A model, for example, is schematically organized, but is a certain representation or logic, which are not the only functions of schemas. Scripts, on the other hand, are schematic structures that denote sequences of events or behavioral procedures,

and categories are schemas used to identify and order things through membership classifications. In other words, models, scripts, and categories are to schemas what fiction or poems are to literature: more specific subcategories.

Culture-Cognition Interaction

This final section discusses how we may think of culture-cognition interaction in a multi-level analytical framework where cultural schemas are treated as macro-level analogues to cognitive schemas. As a starting point for such a framework, I propose a model of culture-cognition interaction based on established models of micro-macro and agency-structure interaction (e.g., Bourdieu 1990; Giddens 1984; Hedström and Bearman 2011). These models describe an interdependent but causally asymmetrical relationship between micro- and macro-phenomena, where macro-structures emerge from micro-level interactions, while in turn structuring future micro-level actions. As illustrated in Figure 8, this model can be used to describe how cultural and cognitive schemas interact without attributing causal primacy to either culture or cognition. In such a model they are interdependent, yet in an asymmetric relationship. What is suggested here is, once again, an analytical framework aimed at interdisciplinary compatibility, and not a theory on its own.

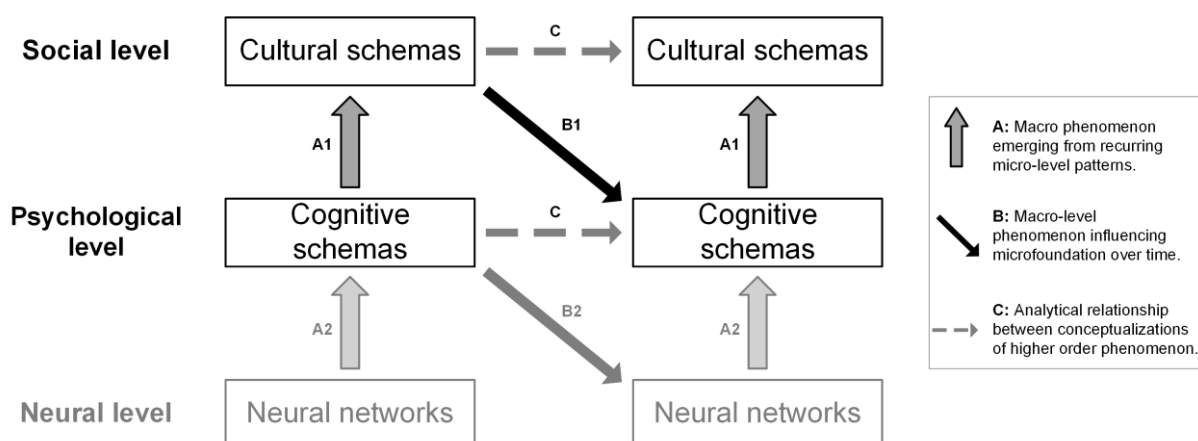


Figure 8: Multilevel analytical framework with emphasis on the asymmetrical relationships of influence between culture and cognition (inspired by Hedström and Bearman 2011)

The model in Figure 8 differentiates between two types of relationships: emergent relationships (A) and causal relationships (B). This distinction emphasizes the asymmetry between culture as a large-scale social phenomenon and cognition as an individual phenomenon. The different relationships between the two also mean that the ways in which one influences the other differs significantly: cultural schemas are beyond the control of individual minds and appear out of necessity when many similar cognitive schemas are enacted, while cognitive schemas are inevitably shaped by surrounding cultural schemas over time.

Reproduction Through Instantiations (A1)

In relationship A1 of Figure 8, cultural schemas emerge from a succession of similar instantiations of cognitive schemas enacted by multiple actors. Because cultural schemas are nothing but a continuity of *patterns* across multiple situations, there is no clear moment when a cultural schema is 'created,' and cultural schemas are never actually stabilized (c.f. Sewell 2005). Instantiations here refer to enactments, or externalizations, of cognitive schemas in the world through actions, articulations, or the shaping of bodies or objects. In contrast to the cognitive schemas themselves, however, instantiations are *observable* by others and, therefore, are the basis of cultural reproduction, or 'transmission,' which in turn leads to the establishment of similar cognitive schemas in other individuals (c.f. Ridgeway 2006). For instantiations to become cultural schemas, they must be similar enough to give rise to overlapping patterns for people to observe or infer, as cultural schemas are not produced by any single individual or action but rather appear as a *social* pattern if sufficiently recurring across different instantiations, individuals, and/or contexts.

To give an example of cultural schemas enacted through instantiations of individuals with similar cognitive schemas learned from countless observations of previous instantiations, we can think of TV news presentations. A news presentation is a highly schematic event where easily recognizable cultural schemas are reproduced as multiple people coordinate various interrelated instantiations of similar cognitive schemas. These include, among other things, roles with associated behavioral scripts, such as the presenter being well-dressed and looking

into the camera, the way the news is presented in a sequential order with a neutral tone, the structuring of the physical environment, usually with a desk and some form of representative background, the coordinated actions of the TV news team, and so forth.

Cultural 'individualists' would object to this characterization of cultural reproduction by saying that what we actually see here is just multiple individuals interacting. That is, an individual-to-individual relationship of influence without any mysterious 'collective objects.' In this, Turner is of course not ontologically wrong, there are just people interacting with people, but the point here is that these people in effect reproduces, and respond to, a supra-individual pattern. It does not matter to Jim's learning process that to him unknown, yet concrete, individuals such as Sarah and Anne are dressing up this particular news presenter for this particular broadcast. Of course, those actions of those concrete individuals are the micro-mechanisms of this particular reproduction, but the important outcome effect for a cultural analyst—and for Jim's learning, thoughts and behavior—is that there exists a strong recurring social pattern, a cultural schema, of how news presentations work, which is maintained through multiple compatible, highly similar, cognitive schemas.

In addition to making it easier to identify and talk about macro-level phenomenon, a separate supra-individual cultural schema concept also emphasizes the process by which individual cognitive schemas are transformed into social phenomenon with emergent properties. Cultural schemas differ significantly from individual cognitive schemas because they are an aggregate (but not sum) of instantiations, and because all instantiations of cognitive schemas are transformative. When cultural schemas emerge from instantiations, they are the product of is a trifold transformative process as:

- i) cognitive schemas are constrained by the ability of the actor to enact abstract schemas as concrete observable instantiations, that is to transform them into concrete actions, articulations or objects, such as for example the limits of language to express an intuitive schema,
- ii) any instantiation of a cognitive schema is constrained by the physical and social environment, including for example the availability of necessary material resources or the presence social control mechanisms such as self-disciplining, norm sanctions, and peer pressure, and

- iii) social processes that determine what instantiations, and in what amount, are seen by who, influenced by for example social power relations, distributions of resources and opportunities, and technology such as social media.

Because of these transformative mechanisms, the resulting cultural schemas are never carbon copies of cognitive schemas. Cultural schemas are not simply the aggravated sum of all instantiations, but emerge from the pattern of overlapping aspects of multiple instantiations. Consequently, because no cultural schema is derived from a specific instantiation, they may differ significantly from any of the individual cognitive schema involved in representing and reproducing them (c.f. Sharifian 2003). Furthermore, as we shall see, internalization of cultural schemas is also a transformative process.

Internalization Through Cognitive Learning (B1)

The second transformative relationship (B1 in Figure 8) describes a causal relationship wherein individual minds are shaped by exposure to culture over time. Cognitive schemas are formed through experiencing socially recurring instantiations of cultural schemas and deriving an abstract pattern from the commonly co-occurring features of those instantiations. This in turn shapes neural networks, as the cognitive representation becomes more frequently used to interpret experiences. So far most would agree on how culture influences individual cognition. Some ‘individualists’ such as Turner (2007b) argue that because people can only learn from observable instantiations, and because of transformative processes, people cannot really attain similar underlying cognitive structures (such as similar presuppositions) when they learn through experiences, but only similar concrete practices. However, there are good reasons to believe that people can infer underlying cognitive structures from observable instantiations.

Human children seem to be hard-wired to perceive, learn, and reproduce cultural patterns in their social environment, because they are predisposed to overimitate adult actions (Nielsen and Tomaselli 2010). Children do this even if the observed pattern has no apparent practical application, a uniquely human trait that is not seen in primates, who imitate only insofar as the practical consequences are apparent. This suggests that children learn not only how to do things, but also social rules, norms and assumptions. As Lizardo (2007) argues, there is good reason to

believe that the so-called Mirror Neuron System is used to infer other people's intentions and reasons, and neuroscientific evidence suggest that such inferences of underlying logics behind people's actions is a low-effort and routine cognitive background operation (Gallese 2003a, 2003b; Meltzoff 2002). Turner's (2007b) argument resembles the 'poverty of stimulus' argument against connectionist networks, which claimed that people cannot infer underlying abstract grammatical rules from overt articulations, but research has shown that people are able to infer complex abstract information from concrete stimuli (see for example Evans 2015 and Shokouhi & Moghimi 2015).

Finally, it should be said that people do not simply learn directly from specific individuals even though those are the micro-mechanism of cultural 'transmission.' What people experiences as cultural schemas 'existing out there' receive an emergent power to influence individual minds, far beyond the power of most individual instantiations, since people often rely on social verification as a feedback mechanism as "experience is established as valid and reliable to the extent that it is shared with others" (Hardin and Higgins 1996:28). People may infer a sense of truth, objectivity, or common sense from what is experienced as something 'everyone just knows' (c.f. Berger and Luckmann 1966). For example, most people are convinced that the Earth is round, not out of personal experiences, but due to the convictions of countless others who together reproduce an unopposed cultural schema of the shape of the Earth. This schema is so frequently reproduced in so many contexts by so many sources that it becomes learned much like a frequently recurring personal experience, which eventually becomes an implicit presumption and an automatic processing mechanism. Even though culture is dependent on cognition, this asymmetry of power must be recognized, as cultural schemas can shape hundreds of thousands, or even millions, of individual minds.

Despite the presence of efficient learning mechanisms, Turner (2007b) is right in that internalization cannot produce exact carbon copies of cultural schemas, because cognitive internalization is not a passive process—where people are invariably socialized in a Parsonian sense—but an active, dynamic and transformative process. All learning occurs continuously through ongoing pattern recognition where people perceive, interpret and respond to experiences by a process of dynamic cognitive (re)construction of schemas corresponding to the experience. As previously described, people use previously learned modular and

transposable cognitive schemas from their own individual learning histories as frames of reference and building blocks to piece together a schema matching their experiences ‘on the go’ (Lizardo 2015; Ridgeway 2006; Smith 1996). This dynamic process using transposable modular schemas also explains how people can identify and respond to new situations in flexible and creative ways which leaves room for variation and social change (see Sewell 2005).

Although patterns are implied in the organization of observable features in concrete instantiations, people also add unobserved elements to their interpretations, such as assumptions, intuitions, and abstract meanings associated with the experienced pattern. A pattern experienced is therefore not simply a sensory experience, but a conjunction of i) previously learned cultural knowledge; ii) the observed organization of sensory elements; and iii) the individual cognitive construction process bringing these together. Because of this dynamic internalization process, individuals will be influenced in different ways by the same experiences due to how their unique learning histories have shaped their neural setup towards certain schematic patterns. Additional factors beyond frequency also influence the impact of experiences on cognitive schemas, such as resonance with previous experiences, emotional arousal (Quinn and Mathews 2016), and rhetorical strength (e.g., discursive power; c.f. Laclau and Mouffe 1985). A single emotionally charged experience, such as a trauma, can sometimes make a memory or interpretative schema highly salient and lasting for life, and claims by authoritative figures such as scientists or priests may more readily be processed as facts than claims by, for example, a child.

Much work remains to be done on internalization processes and how cognitive (re)construction of schemas transforms social patterns into cognitive representations. This paper only has space enough to argue that significant transformation occurs through multiple mediating social and psychological mechanisms. The transformations involved in internalization and reproduction mean that reproducing cultural schemas always occurs through concrete instantiations, in practices and individual cognition, and that this reproduction goes through two transformations—first, from abstract internal schema to observable instantiation, and then back to abstract internal schema through pattern recognition. This suggests that we would benefit from a multi-level analytical framework that distinguishes between social and mental patterns, or cultural and cognitive schemas, as two distinct yet closely connected phenomena.

Conclusions

This paper has argued for the importance of integrating research about cognitive schemas in cultural sociology, and the utility of using cultural schemas as a concept bridging the vocabularies of cognitive science and cultural sociology. The central argument is that it would be most productive to think of cultural schemas as a conceptual analogue for cognitive schemas, retaining a similar conceptual role and structure, but designating a social rather than a mental pattern. This provides us with an accessible language with two-way compatibility which recognizes cognition as the microfoundation of culture, yet maintains culture as a research object in own right, integrates established conceptualizations of culture and facilitates interdisciplinary communication.

The alternative, to use cultural schemas as a subtype of cognitive schemas, leaves us without a way to integrate established cultural theory with cognitive science, which maintains the chasm between the two vocabularies and the fragmentation of cultural theories. A reductionist conceptualization may furthermore lead to continued alienation of most cultural sociologists, which prevents cognitive sociology from being “larger and more inclusive” (Danna 2014:1005). If we want to avoid remaining an “overly insulated ‘gated’ intellectual community” (Cerulo 2010:116), it is in our best interest to make our conceptualizations attractive to others, to avoid ‘psychophobia’ by showing that integration of culture and cognition is possible, and to make good use of insights from established cultural theory. While it is in our best interest that we embrace and adapt to the insights from cognitive science about how the brain processes and reproduces social patterns, there is no inherent contradiction in doing so while treating culture as an emergent supra-individual phenomenon (see Kaidesoja 2013 for a discussion of the biases of individualist microfoundationalism).

We could possibly use cultural schemas as a subtype of cognitive schemas and use another alternative conceptualization for social patterns, but there are significant drawbacks to this strategy. Most cognitive schemas are already inherently cultural and other conceptualizations that could be used for macro-level cultural patterns—such as models, scripts, or categories—are problematic for bridging purposes because they denote specific types of cultural patterns

and therefore are not directly compatible with the schema concept in cognitive science. This compatibility problem is of course also true for discourses, norms, or practices, which refers to types of cultural patterns and not cultural units per se, but this is precisely why we would benefit from using cultural schemas as an integrative umbrella concept much like the cognitive schema.

Using cultural schemas in the way proposed here also has analytical value for an integrated multi-level analytical framework, such as the one sketched out in Figure 8. The model presented here shows that theoretical and analytical integration of knowledge into a vertical model, extending from neural patterns to culture, is possible using an integrated language. This model has the advantages of easily translating schematic patterns, and thus research results, from one level to another. It also emphasizes the asymmetrical relationships between culture and cognition, something which tends to be lost in reductionist frameworks.

Much work remains to be done to translate insights from the cognitive sciences and to construct detailed models of culture-cognition interaction. We need to develop integrated frameworks for dynamic internalization through cognitive construction (c.f. Oyserman 2015; Ridgeway 2006) and dual-process models of culture (c.f. Vaisey 2009). These two areas stand out for their potential to describe in detail the mechanisms of central cultural processes such as learning, reproduction, and how culture shapes the human brain in divergent ways.

References

- Abbe, A., Rentsch, J., & Mot, I. (2009). Cultural Schema: Mental Models Guiding Behavior in a Foreign Culture.
- Alexander, J. C. (2003). *The Meanings of Social Life: A Cultural Sociology*. New York: Oxford University Press.
- Bargh, J. (1997). The Automaticity of Everyday Life. *Advances in social cognition*, 10, 1–61.
- Bargh, J. A., Chen, M., & Burrows, L. (1996). Automaticity of Social Behavior: Direct Effects of Trait Construct and Stereotype-Activation on Action. *Journal of personality and social psychology*, 71(2), 230–44.
- Bartlett, F. C. (1938). *Remembering: A Study in Experimental and Social Psychology*. Cambridge: Cambridge University Press.
- Baumann, S., & Ho, L. (2014). Cultural Schemas for Racial Identity in Canadian Television Advertising. *Canadian Review of Sociology*, 51(2), 152–69.
- Baumann, S., & de Laat, K. (2012). Socially Defunct: A Comparative Analysis of the Underrepresentation of Older Women in Advertising. *Poetics* 40(6), 514–41.
- Bayer, J. B., Campbell, S. W., & Ling, R. (2015). Connection Cues: Activating the Norms and Habits of Social Connectedness. *Communication Theory*, 1–22.
- Berger, P. L., & Luckmann, T. (1966). *The Social Construction of Reality*.
- Blair-Loy, M. (2001). Cultural Constructions of Family Schemas: The Case of Women Finance Executives. *Gender & Society*, 15(5), 687–709.
- Bourdieu, P. (1977). *Outline of a Theory of Practice*. Cambridge University Press.
- Bourdieu, P. (1990). *The Logic of Practice*. Cambridge: Polity Press.
- Brante, T. (2001). Consequences of Realism for Sociological Theory-Building. *Journal for the Theory of Social Behaviour*, 31(2), 167–95.

Brubaker, R., Loveman, M., & Stamatov, P. (2004). Ethnicity as Cognition. *Theory and Society* 33(1), 31–64.

Cerulo, K. (2002). *Culture in Mind: Toward a Sociology of Culture and Cognition*. New York & London: Routledge.

Cerulo, K. (2010). Mining the Intersections of Cognitive Sociology and Neuroscience. *Poetics* 38(2), 115–32.

Cerulo, K. (2014a). Continuing the Story: Maximizing the Intersections of Cognitive Science and Sociology. *Sociological Forum* 29(4), 1012–19.

Cerulo, K., ed. (2014b). *Sociological Forum*. 29(4), 775–1042.

Cerulo, K., & Ruane, J. (2014). Apologies of the Rich and Famous: Cultural, Cognitive, and Social Explanations of Why We Care and Why We Forgive. *Social Psychology Quarterly*, 77(2), 123–49.

Chiao, J. Y. et al. (2010). Theory and Methods in Cultural Neuroscience. *Social cognitive and affective neuroscience*, 5(2-3), 356–61.

Clark, A. (1998). *Being There: Putting Brain, Body, and World Together Again*. Cambridge: The MIT press.

Clark, K. B., & Clark M. P. (1950). Emotional Factors in Racial Identification and Preference in Negro Children. *Journal of Negro Education*, 19(3), 341–50.

Collins, A. M., & Loftus, E. F. (1975). A Spreading-Activation Theory of Semantic Processing. *Readings in Cognitive Science: A Perspective from Psychology and Artificial Intelligence*, 82(6), 126–36.

Conrey, F. R., & Smith, E. R. (2007). Attitude Representation: Attitudes as Patterns in a Distributed, Connectionist Representational System. *Social Cognition*, 25(5), 718–35.

D'Andrade, R. G. (1992). Human motives and cultural models, 23–44.

D'Andrade, R. G. (1995). The Development of Cognitive Anthropology, 122–49.

Danna, K. (2014). The Study of Culture and Cognition. *Sociological Forum*, 29(4), 1001–6.

DiMaggio, P. (1997). Culture and Cognition. *Annual Review of Sociology*, 23(25), 263–87.

DiMaggio, P. (2002). Why Cognitive (and Cultural) Sociology Needs Cognitive Psychology. In K. Cerulo (ed.), *Culture in Mind: Toward a Sociology of Culture and Cognition* (pp. 274–82). New York & London: Routledge.

Evans, G. N. (2015). Two Projects In Theoretical Neuroscience : A Convolution-Based Metric For Neural Membrane Potentials And A Combinatorial Connectionist Semantic Network Method. Penn State University.

Fallin Hunzaker, M. B. (2014). Making Sense of Misfortune Cultural Schemas, Victim Redefinition, and the Perpetuation of Stereotypes. *Social Psychology Quarterly*, 77(2), 166–84.

Foucault, M. (1972). *The Archaeology of Knowledge*. Tavistock Publications.

Frye, M. (2013). Bright Futures in Malawi's New Dawn: Educational Aspirations as Assertions of Identity. *American Journal of Sociology*, 18(9), 1199–1216.

Gallese, V. (2003a). A Neuroscientific Grasp of Concepts: From Control to Representation. *Philosophical transactions of the Royal Society of London. Series B: Biological sciences*, 358, 1231–40.

Gallese, V. (2003b). The Manifold Nature of Interpersonal Relations: The Quest for a Common Mechanism. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 358, 517–28.

Giddens, A. (1984). *The Constitution of Society: Outline of the Theory of Structuration*. Cambridge: Polity Press.

Goffman, E. (1974). *Frame Analysis: An Essay on the Organization of Experience*. London: Harper and Row.

Goodenough, W. (1957). Cultural Anthropology and Linguistics. In P. Garvin (ed.), *Report of the Seventh Annual Round table Meeting on Linguistics and Language Study* (pp. 167–73). Washington D.C.

Gutman, L. M., & Schoon, I. (2013). The Impact of Non-Cognitive Skills on Outcomes for Young People: Literature Review. London.

Hardin, C. D., & Higgins, E. T. (1996). Shared Reality: How Social Verification Makes the Subjective Objective. In R. M. Sorrentino & E. T. Higgins (eds), *Handbook of motivation and cognition, Vol 3: The interpersonal context*.

Hedström, P., & Bearman, P. (2011). What Is Analytical Sociology All About? An Introductory Essay. In *The Oxford Handbook of Analytical Sociology*.

Johnson-Hanks, J. A., Bachrach, C. A., Morgan, S. P., & Kohler, H. (2011). The Theory of Conjunctural Action. In *Understanding Family Change and Variation* (pp. 1–22).

Kaidesoja, T. (2013). Overcoming the Biases of Microfoundationalism: Social Mechanisms and Collective Agents. *Philosophy of the Social Sciences*, 43(3), 301–22.

Laclau, E., & Mouffe C. (1985). *Hegemony and Socialist Strategy*. (2nd ed). New York: Verso Books.

Leys, R. (2011). The Turn to Affect: A Critique. *Critical Inquiry*, 37(3), 434–72.

Lizardo, O. (2007). Mirror Neurons' Collective Objects and the Problem of Transmission: Reconsidering Stephen Turner's Critique of Practice Theory. *Journal for the theory of social behaviour*, 37(3), 319–50.

Lizardo, O. (2014). Beyond the Comtean Schema: The Sociology of Culture and Cognition Versus Cognitive Social Science. *Sociological Forum*, 29(4), 983–89.

Lizardo, O. (2015). Culture, Cognition and Embodiment. *International Encyclopedia of the Social & Behavioral Sciences* 5, 576–81.

Lizardo, O., & Strand, M. (2010). Skills, Toolkits, Contexts and Institutions: Clarifying the Relationship between Different Approaches to Cognition in Cultural Sociology. *Poetics*, 38(2), 205–28.

Mandler, J. M., (1984). *Stories, Scripts, and Scenes: Aspects of Schema Theory*. Hillsdale: Lawrence Erlbaum Associates.

Martin, J. L., & Desmond, M. (2010). Political Position and Social Knowledge. *Sociological Forum*, 25(1), 1–26.

McClelland, J. L., Rumelhart, D. E., & McClelland, J. L. (1986). *Parallel Distributed Processing: Explorations in the Microstructure of Cognition, Volume 2: Psychological and Biological Models*. Cambridge: MIT Press.

Meltzoff, A. N. (2002). Elements of a Developmental Theory of Imitation. In A. N. Meltzoff & W. Prinz (eds.), *The Imitative Mind Development, Evolution and Brain Bases* (pp. 19–41). Cambridge: Cambridge University Press.

Merriam Webster. (2015). Culture. Retrieved December 4, 2015 from <http://www.merriam-webster.com/dictionary/culture>.

Miyamoto, Y, Nisbett, R. E., & Masuda, T. (2006). Culture and the Physical Environment Holistic versus Analytic Perceptual Affordances. *Psychological Science*, 17(2), 113–19.

Nielsen, M., & Tomaselli, K. (2010). Overimitation in Kalahari Bushman Children and the Origins of Human Cultural Cognition. *Psychological science*, 21(5), 729–36.

Nishida, H. (2005). Cultural Schema Theory. In W. B. Gudykunst (ed.), *Theorizing about intercultural communication* (pp. 401–18). California: California State University.

Oyserman, D., Kemmelmeier, M., & Coon, H. (2002). Cultural Psychology, a New Look. *Psychological Bulletin*, 128(1), 110–17.

Oyserman, D. (2015). Culture as Situated Cognition. In R. Scott and S. Kosslyn (eds.), *Emerging Trends in the Social and Behavioral Sciences* (pp. 1–20). John Wiley & Sons.

Parsons, T. (1951). *The Social System*. London: Routledge.

Pitts-Taylor, V. (2014). Cautionary Notes on Navigating the Neurocognitive Turn. *Sociological Forum*, 29(4), 995–1000.

Quinn, N., & Holland, D. (1987). *Cultural Models in Language and Thought*. Cambridge University Press.

Quinn, N., & Mathews, H. F. (2016). Cultural Selves. *Anthropological Theory*, 16(4), 359-389.

Quinn, N., & Strauss, C. (1997). *A Cognitive Theory of Cultural Meaning*. Cambridge: Cambridge University Press.

Ridgeway, C. L. (2006). Linking Social Structure and Interpersonal Behavior: A Theoretical Perspective on Cultural Schemas and Social Relations. *Social Psychology Quarterly*, 69(1), 5–16.

Rosch, E. H. (1973). Natural Categories. *Cognitive Psychology*, 4(3), 328–50.

Schank, R. C., & Abelson, R. P. (1975). Scripts, Plans, and Knowledge. Proceedings of the 4th International Joint Conference on Artificial Intelligence, 151–57.

Sewell, W. (1992). A Theory of Structure. *American Journal of Sociology*, 98(1), 20–24.

Sewell, W. (2005). *Logics of History: Social Theory and Social Transformation*. Chicago: Chicago University Press.

Sharifian, F. (2003). On Cultural Conceptualisations. *Journal of Cognition and Culture*, 3(3), 187–207.

Shepherd, H. (2011). The Cultural Context of Cognition: What the Implicit Association Test Tells Us About How Culture Works. *Sociological Forum*, 26(1), 121–43.

Shokouhi, M., & Moghimi, S. A. (2015). Poverty of Stimulus: A Detrimental or Facilitative Argument in Favor of an Incremental Connectionist Model. *Journal of Applied Linguistics and Language*, 2(3), 111–19.

Smith, E. R. (1996). What Do Connectionism and Social Psychology Offer Each Other? *Journal of personality and social psychology*, 70(5), 893–912.

Smith, E. R. (1998). Mental Representation and Memory. In S. T. Fiske, D. T. Gilbert, & G. Lindzey, *Handbook of social psychology* (pp. 391–445). New York: McGraw-Hill.

Strydom, P. (2007). Introduction: A Cartography of Contemporary Cognitive Social Theory. *European Journal of Social Theory*, 10(3), 339–56.

Swidler, A. (1986). Culture in Action: Symbols and Strategies. *American Sociological Review*, 51(2), 273–86.

Turner, S. P. (2007a). Social Theory as a Cognitive Neuroscience. *European Journal of Social Theory*, 10(3), 357–74.

Turner, S. P. (2007b). Mirror Neurons and Practices: A Response to Lizardo. *Journal for the Theory of Social Behaviour*, 37(3), 351–71.

Vaisey, S. (2009). Motivation and Justification: A Dual-Process Model of Culture in Action. *American Journal of Sociology*, 114(6), 1675–1715.

Valsiner, J. (2013). *An Invitation to Cultural Psychology*. SAGE Publications Ltd.